User Manual

PurityNow Kit for Early E. coli Bacteria Detection in Water

Contents

1. **What Do We Offer?** 3
2. **Materials And Devices:** 3
3. **PurityNow Kit Process Flow:** 4
4. **Procedure:** 5
5. **Precautions:** 6
6. **Technical Details:** 6

Microscopic camera: 6

Battery Used: 6

Temperature Controller: 7

Mobile Application and Server: 7

THE PURITYNOW LEGACY

Dear Customer:

Congratulations on the purchase of PurityNow Kit (Early Detection of Bacterial Contamination in Drinking Water). By investing in PurityNow kit, now you don't need to wait for 24 Hours to get the results of your testing water sample. This kit is specifically design to test E. coli bacterial contamination in drinking water in 7 to 8 hours. E. coli is an indicator of fecal pollution and the possible presence of pathogens.

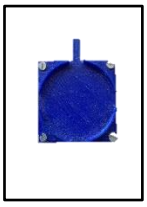
We are committed to making your buying experience as trouble free as possible. You can order by phone, email or via our website which shows images, product specifications and up-to-date stock information 24/7.

We recommend you to read the User Manual carefully before commencing to use the product. Should you require any assistance you can be rest assured that we are just a call away.

1. What Do We Offer?

PurityNow offers cutting-edge technology that enables rapid detection of E. coli bacterial contamination in drinking water through a microscopic camera and artificial intelligence (Ai). With PurityNow you can trust in the accuracy and reliability of your test results.

2. Materials And Devices:


Petri tray


MLSB media


Petri dishes


Forceps


Absorbent pad dispenser


Methanol


Membrane filters


Microscopic camera


Lighter


Disposable gloves


Micropipettes


Absorbent pads



Container for Media preparation



Vacuum pump



Filtration unit



Incubator

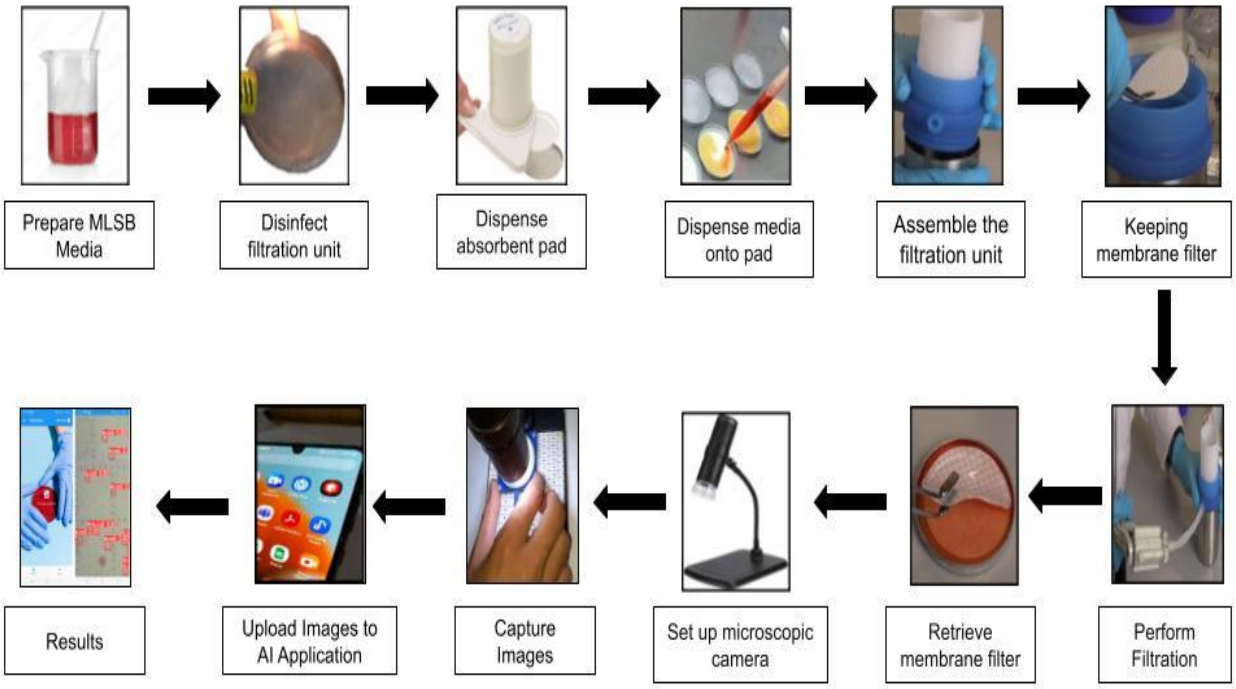


Charger

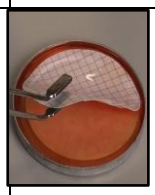
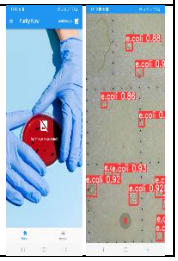


Mobile application

3. PurityNow Kit Process Flow:



4. Procedure:

1. Prepare the MLSB media solution by dissolving 0.762 g in 10 ml of distilled or autoclaved water. Vigorously shake the solution until it turns red.		2. Disinfect the filtration unit and petri dish by applying a small quantity of methanol and subsequently igniting it with a flame and leaving it for a few minutes until it cools down.	
3. Using an absorbent pad dispenser, place an absorbent pad within the petri dish.		4. Using a micropipette, dispense 2 ml of the prepared media evenly onto the absorbent pad within the petri dish.	
5. Assemble the filtration unit.		6. Place a membrane filter onto it using clean forceps.	
7. Measure 100 ml of the water sample and pour it onto the upper portion of the filtration unit.		8. Connect the vacuum pump to the filtration unit and perform filtration, ensuring that the water passes through the membrane filter.	
9. Take out the membrane filter carefully through clean forceps and keep it on the absorbent pad in the petri dish.		10. Note the time and transfer the petri dish to the incubator for a duration of 7 to 8 hours.	
11. Set up a microscopic camera system and connect it to a mobile device.		12. Open the nExt camera mobile application and take images following the pattern defined on the base of the microscopic camera.	
13. Open the PurityNow Mobile application and upload it to the server on which the AI application is running for processing the uploaded images.		14. After few second you will see the processed images back on the screen.	

5. Precautions:

- Sterilize all instruments, equipment, and work surfaces before starting the experiment.
- Wear gloves and protection before experiment.
- Handle water samples with care to prevent cross-contamination.
- Pipets and other vitals should not circulate among working units.
- Handle membrane filters and absorbent pads with sanitized forceps to avoid contamination.
- Regularly monitor the temperature settings to ensure accurate incubation conditions.
- Do not use the media after its expiration date.
- Maintain appropriate temperature in the incubator to support E. coli growth.
- Ensure hand sanitization after performing the experiment.

6. Technical Details:

Microscopic camera:

I have resolution portable USB/wireless microscopic camera is used in the kit which is directly connected to mobile application/edge devices. Other specification is given in the table below.

Network Standard	IEE 802.11 b/g/n
Image sensor	High performance sensor
Sensor size	1/4
Magnification	50-1000 times
Lens structure	2G+IR
Camera light source	8 adjustable LED light
Operating temperature	32 to 113 F
Battery	800mAh lithium battery
Power Input	DC 5V 500 mAh/1A
Focal length	0~infinity
Camera resolution	1920*1080P
Picture format	JPEG
Video format	MP4

Battery Used:

Battery Voltage	12V
Nominal Capacity	25Ah
Dimension	(L) 8.5inches x (W) 2Inches x (H) 2.5Inches
Weight	2.1kg
Battery Technology	Lithium-Ion
Discharge Cut-off Voltage	10.5V
Charge Cut-off Voltage	12.3-12.6V
Operating Voltage Range	10.5 to 12.6V
Protection Circuit Module	Overcharge protection, over-discharge protection, short circuit protection, overcurrent protection

Temperature Controller:

W1209/XH-W1209/HW-557 Digital Thermostat Temperature Controller Incubator Controller Temperature control range -50 to 110°C.

Temperature measurement accuracy	0.1
Control accuracy	0.1
Hysteresis accuracy	0.1
Refresh frequency	0.5 S
Input power	12 V DC

Mobile Application and Server:

Our Ai developed system trained on thousands of images is running online on the server to which access will be provide through a mobile application. It's a very simple user interface and training will be provided to the customer how to efficiently use this system to get the accurate results.



